Rounding and Estimating

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS round *(whakaawhiwhi)* · nearest · tenth · hundredth · decimal place *(mati ā-ira)*

A Rounding whole numbers ●○○

Remember: Look at the digit to the right of the place you are rounding to. If it is 5 or more, round up. If it is 4 or less, round down.

1. Round each number.

a) 47 (nearest 10) _____ **b)** 283 (nearest 10) _____ **c)** 1,365 (nearest 10) _____

d) 840 (nearest 100) _____ **e)** 2,751 (nearest 100) _____ **f)** 6,049 (nearest 100) _____

g) 3,500 (nearest 1,000) _____ **h)** 12,480 (nearest 1,000) _____ **i)** 79,612 (nearest 1,000) _____

j) 45,000 (nearest 10,000) _____ **k)** 128,300 (nearest 10,000) _____ **l)** 996 (nearest 10) _____

B Rounding decimals ●●○

Watch out: Do not just cut the number off. 3.86 to the nearest tenth is 3.9, not 3.8.

2. Round each number.

a) 3.86 (nearest tenth) _____ **b)** 7.4 (nearest whole number) _____

c) 0.273 (nearest hundredth) _____ **d)** 12.65 (nearest tenth) _____

e) 9.5 (nearest whole number) _____ **f)** 4.1082 (nearest thousandth) _____

g) 0.049 (nearest hundredth) _____ **h)** 26.97 (nearest tenth) _____

i) 5.3549 (nearest hundredth) _____ **j)** 0.6 (nearest whole number) _____

k) 8.0364 (nearest thousandth) _____ **l)** 19.96 (nearest tenth) _____

C Decimal places ●●○

Remember: 1 decimal place is the nearest tenth. 2 decimal places is the nearest hundredth. 3 decimal places is the nearest thousandth.

3. Round each number to the number of decimal places shown.

a) 6.283 (1 decimal place) _____ **b)** 0.8465 (2 decimal places) _____ **c)** 14.0729 (3 decimal places) _____

d) 2.55 (1 decimal place) _____ **e)** 31.704 (2 decimal places) _____ **f)** 0.0095 (3 decimal places) _____

g) 7.98 (1 decimal place) _____ **h)** 3.995 (2 decimal places) _____ **i)** 0.3333 (1 decimal place) _____

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D Estimating ●●○

> **Remember:** Round each number so that it has one digit followed by zeros: 38 becomes 40 and 517 becomes 500. Then calculate.

4. Estimate the answer to each calculation. Show the rounded numbers you used.

a) 38 × 21 **b)** 517 + 289 **c)** 62 × 48

d) 792 ÷ 19 **e)** 8.7 × 3.2 **f)** 913 − 388

g) 395 × 52 **h)** 6,120 ÷ 28 **i)** 71 × 9.8

E Is it reasonable? ●●●

> **Remember:** An estimate tells you roughly how big the answer should be. It catches an answer that is ten times too big or too small.

5. One of the three answers is correct. Use an estimate to choose it. Do not work it out exactly.

a) 38×21: 79.8, 798, 7,980

b) $612 \div 18$: 3.4, 34, 340

c) 4.9×6.2: 3.038, 30.38, 303.8

d) $5{,}890 + 3{,}140$: 903, 9,030, 90,300

e) 29×52: 150.8, 1,508, 15,080

f) $7.84 \div 0.98$: 0.8, 8, 80

F Problems ●●●

6. Estimate the answer to each problem. Show the rounded numbers you used.

a) A school orders 28 boxes of pencils. Each box holds 48 pencils. About how many pencils is that?

b) Sam has $19.80. Pies cost $4.95 each. About how many pies can Sam buy?

c) A car travels 589 km on 31 litres of petrol. About how many kilometres does it travel on each litre?

★ Extension

7. A whole number is rounded to the nearest 100 and the answer is 500. What is the smallest the number could be? What is the largest?

8. Round 9.96 to 1 decimal place. Explain why the answer still needs a digit after the decimal point.

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Rounding and Estimating

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS round *(whakaawhiwhi)* · nearest · tenth · hundredth · decimal place *(mati ā-ira)*

A Rounding whole numbers ●○○

Remember: Look at the digit to the right of the place you are rounding to. If it is 5 or more, round up. If it is 4 or less, round down.

1. Round each number.

a) 63 (nearest 10) _____ **b)** 417 (nearest 10) _____ **c)** 2,845 (nearest 10) _____

d) 760 (nearest 100) _____ **e)** 3,249 (nearest 100) _____ **f)** 8,051 (nearest 100) _____

g) 6,500 (nearest 1,000) _____ **h)** 14,520 (nearest 1,000) _____ **i)** 58,397 (nearest 1,000) _____

j) 75,000 (nearest 10,000) _____ **k)** 243,900 (nearest 10,000) _____ **l)** 997 (nearest 10) _____

B Rounding decimals ●●○

Watch out: Do not just cut the number off. 3.86 to the nearest tenth is 3.9, not 3.8.

2. Round each number.

a) 4.73 (nearest tenth) _____ **b)** 8.6 (nearest whole number) _____ **c)** 0.418 (nearest hundredth) _____

d) 15.25 (nearest tenth) _____ **e)** 6.5 (nearest whole number) _____ **f)** 2.3057 (nearest thousandth) _____

g) 0.067 (nearest hundredth) _____ **h)** 34.98 (nearest tenth) _____ **i)** 7.2351 (nearest hundredth) _____

j) 0.4 (nearest whole number) _____ **k)** 5.0716 (nearest thousandth) _____ **l)** 29.95 (nearest tenth) _____

C Decimal places ●●○

Remember: 1 decimal place is the nearest tenth. 2 decimal places is the nearest hundredth. 3 decimal places is the nearest thousandth.

3. Round each number to the number of decimal places shown.

a) 5.172 (1 decimal place) _____ **b)** 0.7385 (2 decimal places) _____ **c)** 11.0468 (3 decimal places) _____

d) 4.45 (1 decimal place) _____ **e)** 26.309 (2 decimal places) _____ **f)** 0.0085 (3 decimal places) _____

g) 6.97 (1 decimal place) _____ **h)** 8.995 (2 decimal places) _____ **i)** 0.6666 (1 decimal place) _____

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D Estimating ●●○

> **Remember:** Round each number so that it has one digit followed by zeros: 38 becomes 40 and 517 becomes 500. Then calculate.

4. Estimate the answer to each calculation. Show the rounded numbers you used.

a) 42 × 29 **b)** 683 + 214 **c)** 57 × 33

d) 612 ÷ 31 **e)** 7.8 × 4.1 **f)** 824 − 479

g) 286 × 41 **h)** 7,930 ÷ 37 **i)** 59 × 6.9

E Is it reasonable? ●●●

> **Remember:** An estimate tells you roughly how big the answer should be. It catches an answer that is ten times too big or too small.

5. One of the three answers is correct. Use an estimate to choose it. Do not work it out exactly.

a) 42×19: 79.8, 798, 7,980

b) $816 \div 24$: 3.4, 34, 340

c) 5.8×7.1: 4.118, 41.18, 411.8

d) $6{,}720 + 2{,}190$: 891, 8,910, 89,100

e) 31×48: 148.8, 1,488, 14,880

f) $6.86 \div 0.98$: 0.7, 7, 70

F Problems ●●●

6. Estimate the answer to each problem. Show the rounded numbers you used.

a) A school orders 32 boxes of pencils. Each box holds 58 pencils. About how many pencils is that?

b) Sam has $29.70. Pies cost $5.95 each. About how many pies can Sam buy?

c) A car travels 812 km on 39 litres of petrol. About how many kilometres does it travel on each litre?

★ Extension

7. A whole number is rounded to the nearest 1,000 and the answer is 7,000. What is the smallest the number could be? What is the largest?

8. Round 19.97 to 1 decimal place. Explain why the answer still needs a digit after the decimal point.

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